

PROJECT PROPOSAL

Web Engineering

Vandalizer

Supervised By

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Course

Web Engineering

Semester

Semester 8

1. Project Team

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2. Introduction & Problem Statement

2.1 Background

Image editing has become an essential part of social media, marketing, design, and everyday digital communication. However, most professional editing tools are still complex for non-expert users. Simple tasks such as removing an object, replacing part of an image, or editing based on a text prompt often require manual selection, masks, layers, and technical knowledge.

Vandalizer is an AI-based image editing web application designed to simplify this process. Instead of hiding the editing workflow behind a single opaque action, the system exposes the major AI stages so the user can understand how an image is detected, segmented, and edited.

2.2 Problem Statement

Many users want to quickly edit an image using natural language, but existing tools are either too manual or too abstract. This project addresses how to provide a simple web interface that allows a user to describe an edit in text while the system automatically detects the target object, segments it, and replaces it using AI-powered inpainting.

2.3 Motivation

The motivation behind Vandalizer is to make AI image editing easier for regular users and more understandable for students and beginners. The project also has educational value because it demonstrates how modern vision-language models, segmentation models, and diffusion models work together in a practical web application.

3. Objectives

By the end of this project, the team will have achieved the following measurable goals:

- Build a web application that accepts an image and a text prompt for editing.
- Automatically detect the object described in the prompt inside the uploaded image.
- Segment the detected object accurately using an AI segmentation model.
- Replace or edit the selected region using a diffusion-based inpainting model.
- Present the editing pipeline in a clear and understandable frontend for simple users.

4. Literature Review

This project is based on several important ideas in modern computer vision and generative AI.

- **Grounded object detection:** Models such as Grounding DINO and OWL-V2 connect text descriptions to image regions and help locate objects using language.
- **Segmentation:** SAM and MobileSAM are used to separate the selected object from the background with pixel-level precision.
- **Transformer architectures:** Transformer-based models are important for combining visual and textual information.
- **Diffusion-based inpainting:** Stable Diffusion XL Inpaint and similar models generate realistic replacements in masked areas.
- **Image restoration methods:** LaMa-style inpainting methods provide useful background for understanding image completion and reconstruction.

5. Proposed Methodology

5.1 System Overview

Vandalizer is a client-server web application. The user uploads an image through the React frontend and enters a text prompt describing the object to edit. The FastAPI backend receives the image, creates a job ID, and stores the file on the server. A Celery worker connected to Redis runs the AI pipeline asynchronously.

The pipeline works in three main stages: object detection from the text prompt, segmentation of the detected object, and inpainting or replacement of the segmented area. The frontend polls the server and displays the detection boxes, segmentation mask, and final edited image.

5.2 Application Components

Component	Description
React Frontend	User interface for upload, prompt entry, status polling, and result display.
FastAPI Backend	Handles upload, job creation, status checking, and process endpoints.
Celery Worker	Executes model inference asynchronously so the interface remains responsive.
Redis	Acts as the message broker and result backend for Celery.
Object Detection Model	Text-grounded detector that finds the target object in the image.
Segmentation Model	MobileSAM or SAM used to generate a precise object mask.
Inpainting Model	Stable Diffusion XL Inpaint used to generate the edited output.

5.3 Software Architecture

The software uses a modular architecture with a separate frontend and backend. The frontend is built with React and Vite, while the backend is built with FastAPI and Celery. Redis supports asynchronous task execution. The AI models are loaded through a service layer so they can be reused efficiently across requests.

This architecture is suitable for separating user interaction from heavy model inference, which improves maintainability and keeps the application responsive.

5.4 System Flow

- The user uploads an image through the frontend.
- The backend saves the image and generates a job ID.
- The user enters a text prompt describing the object to edit.
- The detector identifies the object region using text-guided detection.
- The segmentation model creates a mask for the detected object.
- The inpainting model replaces the masked region using the prompt.
- The frontend polls the server and displays the result when processing is complete.

6. Development Timeline

The project will be completed over 6 to 7 weeks according to the following plan:

Week	Phase / Task	Deliverable
9	Prepare or train the AI model pipeline and test initial inference	Working detection / segmentation / inpainting base
10	Set up the backend with FastAPI, Redis, and Celery	Functional backend APIs and async processing
11	Set up the frontend with React and connect the upload flow	Basic user interface for uploading images and prompts
12	Ensure seamless frontend-backend integration and fine-tune the model pipeline	End-to-end working system
13	Improve the frontend design and make the interface clean and user-friendly	Polished UI
14	Deploy the application and prepare production settings	Deployed version
15	Final testing, bug fixing, and presentation preparation	Final submission